

P (tools/armor) = Refined Plutonium Ingot | R = Radioactive Core | _ = empty slot

Phase 3 -- Radiation Drill (only tool that can mine Plutonium Ore)

D

G

D

R

R

R

D

S

D

D = Diamond (4)

G = Glowstone Dust (1)

R = Radioactive Core* (3)

S = Stick (1)

* Radioactive Core = custom item (PDC-validated Magma Cream)

Phase 5 -- Nuclear Smelter (crafting table)

D

O

D

O

F

O

D

O

D

D = Diamond (4)

O = Obsidian (4)

F = Furnace (1)

Nuclear Smelter -- Machine Processing (place items into GUI, not crafting table)

Input	Output	Processing Time	Fuel/cycle
Raw Plutonium Fragment	Refined Plutonium Ingot	300 ticks (15 s)	1 Core

Phase 6 -- Hazmat Armor (W = Yellow Wool)

Radiation resistance: Helmet +20% | Chestplate +30% | Leggings +20% | Boots +10% -> Full set: 80% reduction

Hazmat Helmet

W

W

W

W

W

5 Yellow Wool

Hazmat Chestplate

W

W

W

W

W

W

W

W

8 Yellow Wool

Hazmat Leggings

W

W

W

W

W

W

W

7 Yellow Wool

Hazmat Boots

W

W

W

W

4 Yellow Wool

Phase 6 -- Plutonium Armor (P = Refined Plutonium Ingot, PDC-validated)

Full set: 100% environmental radiation immunity + Passives: Night Vision, Regen, Speed, Reduced Fall Damage

Plutonium Helmet

P

P

P

P

P

5 ingots

Plutonium Chestplate

P

P

P

P

P

P

P

P

8 ingots

Plutonium Leggings

P

P

P

P

P

P

P

7 ingots

Plutonium Boots

P

P

P

P

4 ingots

Phase 6 -- Plutonium Tools (P = Refined Plutonium Ingot | S = Stick)

Tool	Grid Pattern (row1 row2 row3)	Cost
Plutonium Sword	_P_ _P_ _S_	2P + 1S
Plutonium Axe	PP_ PS_ _S_	3P + 2S
Plutonium Pickaxe	PPP _S_ _S_	3P + 2S
Plutonium Shovel	_P_ _S_ _S_	1P + 2S
Plutonium Hoe	PP_ _S_ _S_	2P + 2S

Phase 6 -- Plutonium Arrow (yields 4 per craft)

	P	
	A	
	F	

- P = Refined Plutonium Ingot (1)
- A = Arrow (1)
- F = Feather (1)
- * On hit: irradiates target, applies Poison + Glowing

Phase 7 -- Radiation Antidote (Shapeless -- any slot arrangement)

2x Healing Petal + 1x Honey Bottle -> 1 Radiation Antidote
Reduces current radiation level. Does NOT grant immunity.

Phase 7 -- Radiation Serum (grants temporary radiation immunity)

P	R	P
H	G	H
P	B	P

- P = Healing Petal (4)
R = Radioactive Core (1)
H = Gold Nugget (2)
G = Golden Apple (1)
B = Glass Bottle (1)

Phase 8 -- Nuclear Forge Block (crafting table)

O	D	O
E	S	E
O	D	O

- O = Obsidian (4)
D = Ancient Debris (2)
E = Echo Shard (2)
S = Smithing Table (1)

* Right-click the placed block to open the upgrade GUI

Phase 8 -- Nuclear Forge Upgrades (machine operation, not crafting table)

Place equipment in Equipment slot | ingots in Material slot | cores+hearts in Catalyst slot | press Forge.
* Energy stored via Radioactive Cores (2,000 energy each, 10,000 max). Energy decays slowly when idle.

Tier	Refined Plut. Ingots	Radioactive Cores	Irradiated Hearts	Energy Cost	Success %
MK-I	2	1	--	500	100%
MK-II	4	2	--	1,200	90%
MK-III	8	4	1	2,500	75%
MK-IV	16	8	2	5,000	50%

(!) MK-IV Warnings:

- 5% chance to downgrade item to MK-III on a failed attempt.
- MK-IV passively emits a Radiation Aura (3-block radius, every 2 sec) -- irradiates nearby hostile mobs and enemy players.

Full MK-IV Cost Summary (one equipment piece, raw materials all the way to MK-IV)

Resource	Breakdown	Minimum Total
Refined Plutonium Ingots	Piece (4-8) + upgrades (2+4+8+16=30)	34-38
Radioactive Cores	Drill (3) + forge energy + upgrades (15)	>=18
Irradiated Hearts	MK-III: 1 + MK-IV: 2 (per success)	3 min.
Forge Energy	500+1,200+2,500+5,000 (successful run)	>=9,200
Ancient Debris	Forge block crafting only	2
Irradiated Heart item	Base material: Heart of the Sea (PDC-tagged)	varies

